

Problem Statement: The Titanic disaster is one of the most famous shipwrecks in history. In this exercise, you will explore the Titanic dataset containing passenger information such as age, sex, passenger class, fare, and whether they survived or not. Your goal is to perform Exploratory Data Analysis (EDA) to uncover patterns and insights that might explain survival chances.

Learning Objectives:

- Load and inspect a real-world dataset
- Understand dataset structure and summary statistics
- Calculate central tendency measures: mean, median, mode
- Identify and visualize missing data
- Perform univariate and bivariate analysis using plots
- Explore correlations between numerical variables
- Gain hands-on experience with pandas and seaborn libraries

Tasks to be completed step-by-step:

- Step1: Import necessary libraries.
- Step2: Load the Titanic dataset and preview it.
- Step3: Check dataset shape, data types, and missing values.
- Step4: Calculate descriptive statistics and central tendency (mean, median, mode).
- Step5: Visualize distributions and key statistics.
- Step6: Analyze missing data and discuss its impact.
- Step7: Perform univariate analysis of individual features.
- Step8: Perform bivariate analysis to study relationships between features.
- Step9: Compute and visualize the correlation matrix for numeric features.
- Step10: Summarize findings and prepare for further modeling or data cleaning.

```
# Step 1: Import Libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# Step 2: Load Titanic Dataset
df = sns.load_dataset('titanic')
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
# Step 3: Dataset Shape and Info
print(f"Dataset shape (rows, columns): {df.shape}\n")
print(df.info())
```

```
Dataset shape (rows, columns): (891, 15)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
```

```

4  sibsp      891 non-null  int64
5  parch      891 non-null  int64
6  fare       891 non-null  float64
7  embarked   889 non-null  object
8  class      891 non-null  category
9  who        891 non-null  object
10 adult_male  891 non-null  bool
11 deck       203 non-null  category
12 embark_town 889 non-null  object
13 alive      891 non-null  object
14 alone      891 non-null  bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None

```

```

# Step 4: Summary Statistics
print(df.describe())

```

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```

# Step 5: Calculate Mean, Median, Mode for 'age'
mean_age = df['age'].mean()
median_age = df['age'].median()
mode_age = df['age'].mode()[0]
print(f"Mean Age: {mean_age:.2f}")
print(f"Median Age: {median_age:.2f}")
print(f"Mode Age: {mode_age}")

```

```

Mean Age: 29.70
Median Age: 28.00
Mode Age: 24.0

```

```

# Mode for categorical 'embarked'
mode_embarked = df['embarked'].mode()[0]
print(f"Most common Embarked Port: {mode_embarked}")

```

```

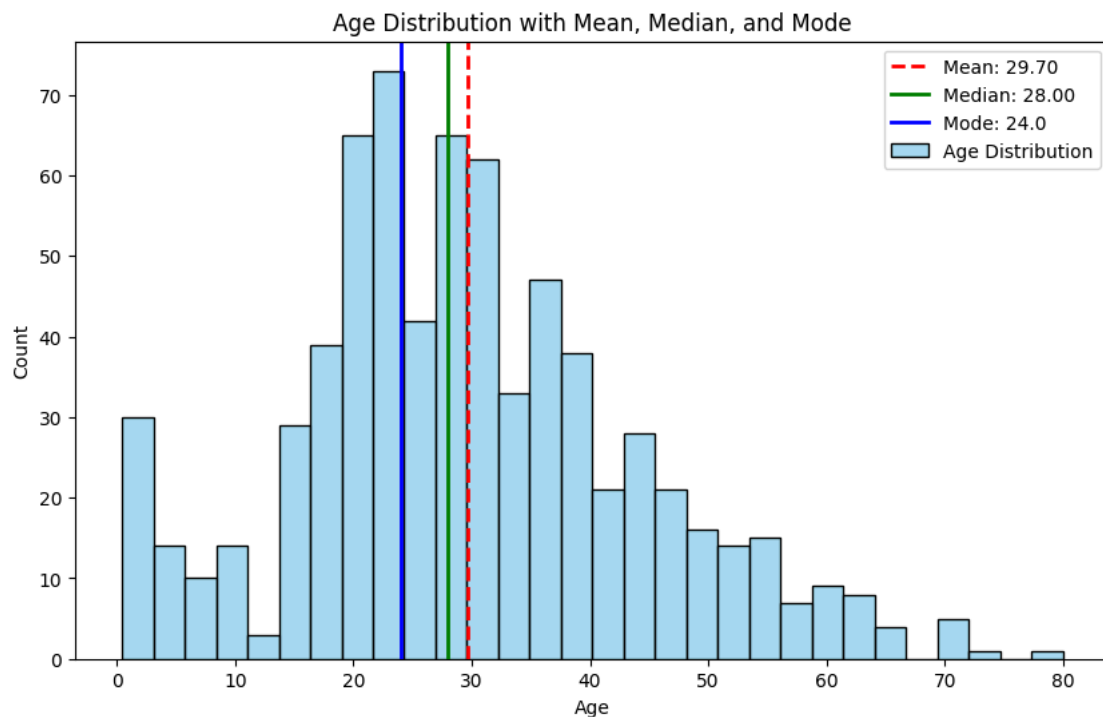
Most common Embarked Port: S

```

```

# Step 6: Visualize Age Distribution with Mean, Median, Mode
plt.figure(figsize=(10,6))
sns.histplot(df['age'].dropna(), bins=30, kde=False, color='skyblue', label='Age Distribution')
plt.axvline(mean_age, color='red', linestyle='--', linewidth=2, label=f'Mean: {mean_age:.2f}')
plt.axvline(median_age, color='green', linestyle='-', linewidth=2, label=f'Median: {median_age:.2f}')
plt.axvline(mode_age, color='blue', linestyle='-', linewidth=2, label=f'Mode: {mode_age}')
plt.title('Age Distribution with Mean, Median, and Mode')
plt.xlabel('Age')
plt.ylabel('Count')
plt.legend()
plt.show()

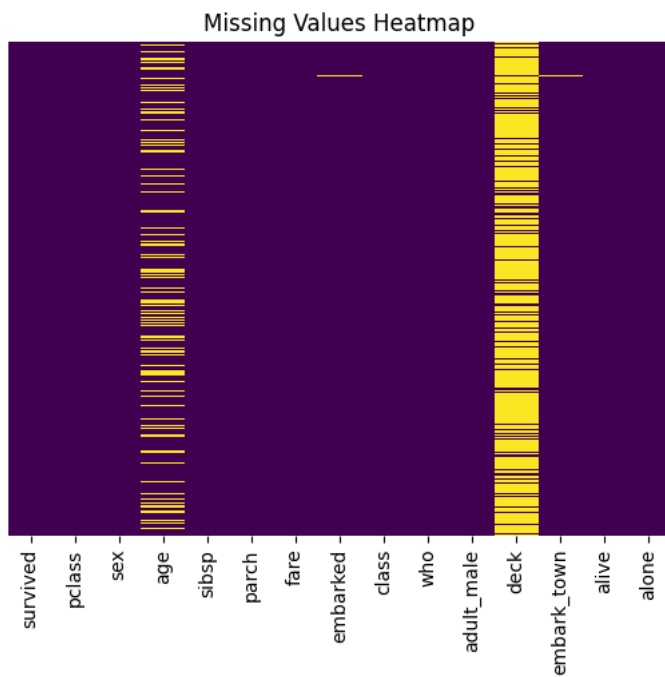
```



```
# Step 7: Check Missing Values  
print(df.isnull().sum())
```

```
survived      0  
pclass        0  
sex           0  
age          177  
sibsp         0  
parch         0  
fare          0  
embarked      2  
class         0  
who           0  
adult_male    0  
deck         688  
embark_town   2  
alive         0  
alone         0  
dtype: int64
```

```
# Visualize missing data  
sns.heatmap(df.isnull(), cbar=False, yticklabels=False, cmap='viridis')  
plt.title('Missing Values Heatmap')  
plt.show()
```

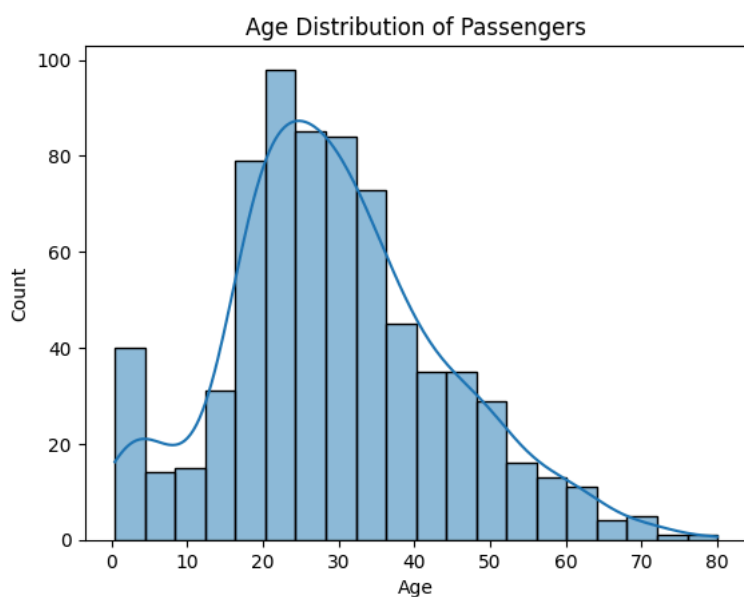
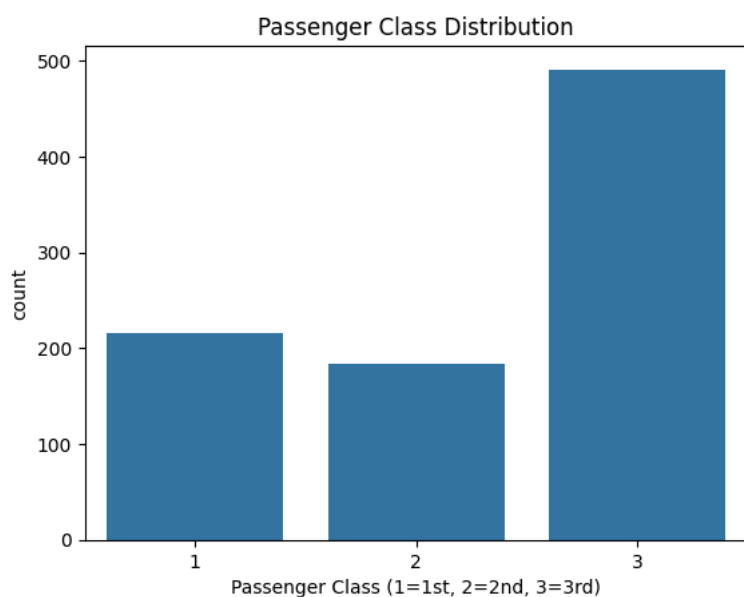
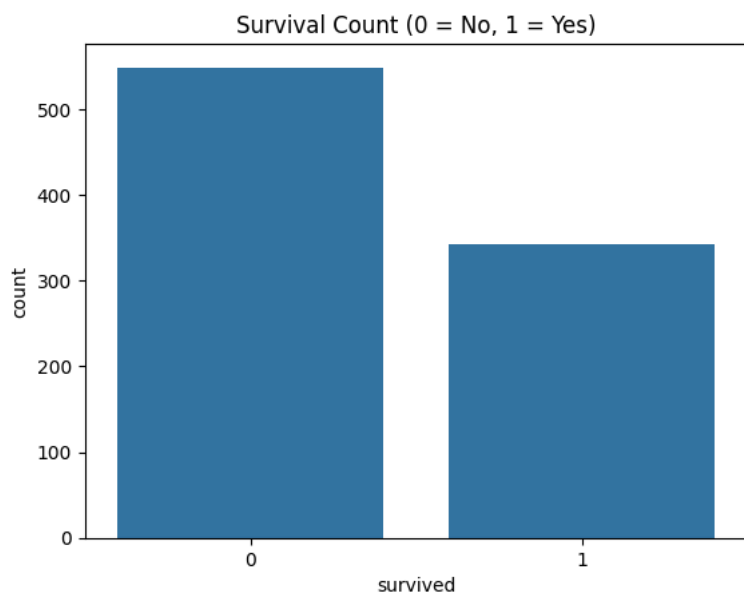


Step 8: Univariate Analysis

```
# Survival count
sns.countplot(x='survived', data=df)
plt.title('Survival Count (0 = No, 1 = Yes)')
plt.show()

# Passenger class distribution
sns.countplot(x='pclass', data=df)
plt.title('Passenger Class Distribution')
plt.xlabel('Passenger Class (1=1st, 2=2nd, 3=3rd)')
plt.show()

# Age distribution
sns.histplot(df['age'].dropna(), kde=True)
plt.title('Age Distribution of Passengers')
plt.xlabel('Age')
plt.show()
```



Step 9: Bivariate Analysis

```
# Survival by gender
sns.countplot(x='survived', hue='sex', data=df)
plt.title('Survival by Gender')
plt.show()
```

```
# Age distribution by survival status
```

```
sns.boxplot(x='survived', y='age', data=df)
plt.title('Age Distribution by Survival Status')
plt.xlabel('Survived (0=No, 1=Yes)')
plt.ylabel('Age')
plt.show()
```

```
# Fare distribution by survival
sns.boxplot(x='survived', y='fare', data=df)
plt.title('Fare Distribution by Survival')
plt.xlabel('Survived (0=No, 1=Yes)')
plt.ylabel('Fare')
```

